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CAREER SUMMARY

Thank you for choosing Ellis Porter PLC's High-Skilled Immigration Team. We've built our practice on the belief that every case is unique and should be treated as such. We want to build a case strategy around your specific background, expertise, and credentials, and this is the document that will help us best accomplish that. We ask that you please fill out each section as completely and thoroughly as possible so that we can best understand your career and what makes you special. If you have any questions or concerns while you work on it, please let us know. We look forward to working with you to prepare the strongest petition possible.

Section A: Your Area of Expertise

In this section, we want to learn about the area in which you work. A few sentences should be sufficient to answer most of these questions.

1. **What do you consider to be your field? This should be a very short description – a few words at most – and should be broad enough to cover all of your past, current, and anticipated future work**:

   Geographic Information Science (GIS)

2. **Please provide a brief summary of the focus of your past work**:
   Research:

   Within the field of GIS, my primary focus is on modeling coastal environments and coastal change due to sea level rise and climate change issues. I perform different hydrodynamic, geomorphological, ecological and societal numerical modeling, and also use them as tools for disseminating information to be used for the prediction of future scenarios.

   For instance, I was a geospatial lead for one of the most comprehensive and multidisciplinary sea level rise assessments ever performed in the upper Louisiana coast titled "Living with sea level rise in Louisiana." For that project, I developed a dynamic modeling framework to model the potential environmental and socioeconomic impacts due to rising sea level and concomitant enhanced storm surge caused by higher sea level and changes in land cover. The same modeling framework is implemented to model the impacts in the whole Louisiana coast for the Louisiana Coastal Resiliency Initiative, a comprehensive statewide plan to protect and promote resilient Louisiana coast by

Louisiana Land Office (LLO). I have also undertaken study to quantify the storm protection service provided by the coastal wetlands in protecting the coastal ecosystems and thus maintaining communities' well-being.

3. **Please provide a brief description of what you hope to do in your future work in the US**:

   I will be continuing the similar research theme that I have been conducting at Higher Research Academy, but will be seeking for scientific and technological improvements. Here are few research topics that I hope to do in the future:

   - Model development: I am interested in the improvement of the currently available coastal marsh evolution models being used to project the impact of sea level rise in the future by incorporating future changes in hydrodynamics that will affect suspended sediment, tide ranges, freshwater inflows, and erosion. There are only a handful of these types of models available, and none of them account for the complex feedbacks between storms, currents, waves, and sediment supply that shape beaches and dunes.
   - Model integration: I will also be working on integrating a couple of other models in the dynamic modeling framework that I have developed to calculate the suspended sediment transport and to estimate freshwater river discharges and saltwater intrusion in bays and estuaries. In fact, I will be working on this soon as we have an upcoming project for the Port of Grand Ilse to model potential changes to storm surge, tides and salinity due to deepening/widening of the Grand Isle Ship Channel.
   - Storm Protection Study: I want to expand a case study that I did to explore the storm protection service provided by coastal wetlands of the Grand Isle Bay System (*manuscript in review*). I want to incorporate multiple storm conditions by simulating a handful of hypothetical storms of different characteristics to get a holistic understanding of the attenuating effect of coastal wetland and quantify the storm protection service provided by the coastal wetland.
   - Website development: We have quickly put together all the results and data of "Living with sea level rise in Louisiana" in a website. We have recently finished the statewide sea level rise and storm surge impact modeling for integration into the website as well. I want to improve user experience for using the website and develop it as a One-Stop decision support system providing all the information regarding the impacts of sea level rise and storm surge in future in the Louisiana coast.

4. **Please provide a brief description of your expertise. What makes you an expert in your field? What separates you from your peers**?

   First of all, I am able to combine a number of technical skills: I have expertise in spatial data analysis, remote sensing, web mapping and different types of coastal numerical modeling like hydrodynamic, geomorphological and ecological modeling. I have developed a dynamic modeling framework assessing sea level rise and climate change effects on coastal inundation by combining hydrodynamic, morphologic and ecological models. In addition to modeling these biophysical impacts, I have also combined another modeling tool in the framework to analyze the socioeconomic impacts due to rising sea level and concomitant enhanced storm surge caused by higher sea level and changes in land cover in the Louisiana coast.

   Because of my diverse academic background, with degrees in Computer Science and Geographical Information Science (GIS), I am uniquely qualified for to understand and develop the underlying spatial analysis and computer programming for coastal applications. My Computer Science degree has provided me with a broad background and knowledge of tools that are really powerful when applied to Geography problems, enabling me to turn the data into information. In essence, I am able to both develop the computer programs that facilitate this research, and to actually carry out the research myself. This is very rare in my field and makes me highly desirable for many research projects.

5. **Please briefly describe the importance of work in your field. Broadly speaking, why is this type of work important**?

   The Louisiana coast is quite vulnerable to sea level rise because of its flat topography, low-lying shorelines and adjacent coastal environments. It is also especially vulnerable to storm surge due to the frequent hurricanes and tropical storms. In addition to those, the sinking of coastal lands has increased the frequency and severity of flooding in the region. Therefore, the work that I have been doing is really important to ensure a promising future along the Louisiana coast. My modeling work provides the baseline information to further comprehend and predict potential future scenarios for sea level rise, coastal storm and flood risk, and infrastructure impacts. Similarly, this type of modeling work illustrates the need and benefit of the natural-based and gray infrastructure into the future to cope with the sea level rise and climate change. This type of work is necessary for effective coastal planning to improve our capabilities to prepare for, resist and adapt to coastal hazards, and increasing the resiliency of our coastal region.

6. **Please briefly describe the impact of your specific work. Who benefits from the work that you do**?

   The projects that I am involved enable policy-makers, resource managers and the general public in the Louisiana coast to better evaluate the impacts or risks of future sea level rise on their communities with greater precision and accuracy. For instance, I have developed a website that provides massive amounts of information to evaluate the environmental, socioeconomic, and legal aspects of sea level rise on our community. The website has photos and maps with detailed narrative to guide everyone, along with an option to download the data to explore on their own. Similarly, my work is beneficial to landowners and developers in the Louisiana coast to evaluate the impacts or risks of private and public land use decisions because sea level rise is going to impact and alter the coastal landscape, both the topography and land cover, along the coastal margins. They are also helpful to inform policy and decision-making to promote conservation and restoration practices that lead to the preservation and enhancement of coastal communities' well-being.a

Section B: Your Academic Achievements

In this section, we want to get a detailed look at your educational history. Please answer the following questions for each degree you have earned. We realize some of this information may be in your CV, but it helps us work most efficiently to have all your information in one place.

1. **Degree earned**: Master of Science (MS) degree
2. **Area of study**: Geographic Information Science for Development and Environment
3. **Institution**: Michigan State University
4. **If known and notable, any rankings of your institution**:
5. **Dates of attendance**: 2012 – 2014
6. **Please list your accomplishments while pursuing this degree. For instance, honors, scholarships, student awards, etc. can be listed here**:

   IDCE Tuition Fellowship
   Best Poster Award at [conference]

1. **Degree earned**: Master of Science (MS) degree
2. **Area of study**: Computer Science
3. **Institution**: University of Michigan
4. **If known and notable, any rankings of your institution**: #25 in National University Rankings, #3 ranked computer science program by US News and World Report
5. **Dates of attendance**: 2008 – 2010
6. **Please list your accomplishments while pursuing this degree. For instance, honors, scholarships, student awards, etc. can be listed here**:

1. **Degree earned**: Bachelor of Engineering (BE) degree
2. **Area of study**: Computer Engineering
3. **Institution**: Pokhara University, Nepal
4. **If known and notable, any rankings of your institution**:
5. **Dates of attendance**: 2001 – 2006
6. **Please list your accomplishments while pursuing this degree. For instance, honors, scholarships, student awards, etc. can be listed here**:

   IOE Score-based Scholarship

Section C: Your Employment History

In this section, we want to get a detailed look at your employment history. Please answer the following questions for each full-time position in your field that you have held. You do not need to include employment from before receiving your Bachelor's degree. We realize some of this information may be in your CV, but it helps us work most efficiently to have all your information in one place.

1. **Employer name**: Louisiana State University
2. **If known and notable, the reputation of your employer**:
3. **Job title**: Research Data Scientist
4. **Detailed job description and list of duties**:
    a. Conduct and manage research project on modeling the effects of sea level rise and land cover change on future storm surge impacts due to tropical cyclone-driven storm along the Louisiana coast for the Louisiana Coastal Resiliency Initiative and `Living with Sea Level Rise in Upper Louisiana Coast' project
    b. Subject Matter Expert for reviewing the dataset submitted files of geospatial numerical models to the Gulf of Mexico Research Initiative Information and Data Cooperative (GRIIDC)
    c. Design decision support and scientific visualization tools. Develop web-based maps and tools.
    d. Publish scientific journal articles. Write proposals, procedures, manuals, reports, and presentations. Attend meeting/conference to make presentations.
    e. Research coastal environments and change. Perform numerical modeling (hydrodynamic/geomorphological/ecological/societal) and spatial data analysis.
    f. Maintain and operate High Performance Computing (HPC) environments.
5. **Dates of employment**: August 2014 - Present
6. **Salary**: $68,300

1. **Employer name**: Louisiana Diversity Internship Program
2. **If known and notable, the reputation of your employer**:
3. **Job title**: Diversity Intern
4. **Detailed job description and list of duties**:
    a. Conducted extensive research related to time-series analysis of status of Cedar Bayou, a natural pass in Louisiana, based on morphologic responses of the pass to past processes using LiDAR, aerial imagery, wave statistics, and other process information
    b. Prepared maps; poster presented on NESTVAL Conference
5. **Dates of employment**: May 2013 – August 2013
6. **Salary**: N/A

Section D: Detailed Summary of Your Work

This is perhaps the most important section in allowing us to understand your work. Here, you should break your career up into different "projects" and provide a description of what you did. A project could be one specific area on which you worked for an extended period of time, a collection of smaller tasks that were related in some way, or a specific step in your career progression. Please try to describe at least 2-3 projects, but no more than 6. In this template, we have included spots for 3 projects; please add additional entries as needed. If you're having trouble deciding how to organize or select your projects, please let us know and we can provide some suggestions about what might be best for you.

1. **PROJECT 1 – Name**: Living with Sea Level Rise on the Upper Louisiana Coast
2. **Date project began**: August 2014
3. **Date project ended**: August 2016
4. **Please list any funding this project received (if applicable)**: $720,000 from Louisiana Endowment
5. **Please list all resulting publications (if applicable)**:

    a. Project deliverable: [website]
    b. "Investigating sea level rise and possible future landscapes on storm surge in the Grand Isle region", Abstract presented at ASBPA Louisiana Chapter 2018
    c. "Sea Level Rise effects and Land Cover Change on Storm Surge Impacts in the Grand Isle, Louisiana region", Abstract presented at 2018 Ocean Sciences Meeting
    d. "The Impact of Coastal Policies Regarding Sea Level Rise in Grand Isle", Restore America's Estuaries 8th National Summit on Coastal and Estuarine Restoration and 25th Biennial Meeting of The Coastal Society, 2016.
    e. "The Ongoing Biophysical Impact of Sea Level Rise Policies in Grand Isle, Louisiana", Abstract presented at Society of Wetland Scientists' 2016 Annual Meeting
    f. "Determining the Environmental and Socioeconomic Impact of Sea Level Rise in the Grand Isle Bay", Abstract presented at 2016 Ocean Sciences Meeting
    g. "Louisiana Coastal Regions: Socio-economic Impact of Sea Level Rise in Grand Isle", Abstract presented at the 8th Annual Hazus User Conference, 2015

6. **Please provide any necessary background information for us to understand this project. What is the significance of this work? Why did you undertake this project? Was there a specific issue or problem you were trying to resolve? What were you hoping to accomplish?**

This project is one of the most comprehensive and multidisciplinary sea level rise assessments ever performed in Louisiana to provide stakeholders with the information they need to understand and adapt to higher sea levels in next 50-100 years. It takes a unique, holistic approach to examine and model potential environmental and human impacts, and also review existing coastal management strategies to understand if they apply to the projected sea level rise in the future. It is a multidisciplinary project as it involves three broad disciplines so three research groups – geospatial, socioeconomic, and policy and law. The geospatial group assessed and modeled the impacts of sea level rise on natural and build environments in the greater Grand Isle area considering three sea level rise scenarios in the future. The socioeconomic group developed and conducted a survey to capture data about the value residents in the greater Grand Isle area place in the local environment. The policy and law group reviewed law/policy currently on the books that could affect potential adaptation strategies for dealing with sea level rise.

The population of the greater Grand Isle area did not have complete information on the extent of projected sea level rise and its social and economic impacts. These information gaps pose a serious threat to the stability of the regions' natural and built environments into the next century and could seriously hinder or delay the regions' social and economic progress. The natural and built environments of the greater Grand Isle area are part of a low-lying coastal plain that is undergoing sea-level rise. This rise will continue to cause critical coastal environments, such as coastal wetlands, to migrate or convert to open water. As a result, the built environment and human wellbeing is becoming increasingly less stable as protective natural environments diminish. This study assesses the growing vulnerability of Grand Isle and its surrounding counties to the adverse consequences of sea level rise by considering three global sea level rise scenarios from 1.8 m in 2100 to 0.21 m in 2100. These scenarios are used in this study to represent a probable lower limit, a moderate scenario, and a probable upper limit of sea level rise by 2100.

Since 1908, the tide gauge at Pier 21 on Grand Isle has recorded a rise in relative sea level of about 2 feet. Roughly 1 foot of this rise is due to a global increase in ocean water volume caused by climate change with the remainder caused by local land sinking called as subsidence. The amount of relative sea-level rise across the greater Grand Isle area varies because of differences in how much the land is subsiding. Therefore, a subsidence rate grid is developed for this project and is used in the sea level rise impact modeling. So, this project assesses future sea level rise impacts caused by local land subsidence and global sea-level rise caused by climate change.

7. **Please provide a plain language summary of what was done in this project. You should try to be detailed (2-4 paragraphs is appropriate), as more information will help us better understand the work**:

The project was funded in September 2013 but while I joined Higher Research Academy (HRA) in August 2014 almost nothing had been done for a year. I quickly learned about the scope of this project, found it interesting and challenging, and took a role as the geospatial lead and manager of the project. As the geospatial lead, I developed the modeling framework to model the biophysical and socioeconomic impacts of sea level rise in both the natural and built environments. As a project manager, I managed the project by coordinating and tracking a team of graduate students and research staffs along with two research groups in HRA.

I added an additional storm surge modeling component to the project although it was not required for the project deliverable. I modeled the impacts of sea level rise and concomitant enhanced storm surge caused by higher sea level and changes in land cover. Simply put, I assess the effects of future landscapes and sea level rise on storm surge by considering previous hurricanes as a representative storm making landfall in 2100 AD and compare it with the impacts of previous hurricanes on present landscapes. This basically is a new capability that I am able to add in our lab which led to additional funding from Louisiana Land Office for the Louisiana Coastal Resiliency Initiative and Coastal Louisiana Study, and just recently from the Port of Grand Isle.

This project provides maps of the current status and ecological services of coastal areas and projects future changes in the distribution of specific environments. By showing areas that are at risk to the negative impacts of sea level rise and estimating the environmental and socioeconomic cost of that impact, project results provide essential information for planning the conservation, protection, or restoration of coastal environments, help guide private and public land acquisitions, and identify new locations for growth and development. This project also provides analysis on the dynamics of coastal environments needed to determine the design, viability, and lifespan of restoration projects in specific locations.

8. **What was your specific role on this project? What were you individually responsible for? How was your work critical to the outcome of the project**?

I was a geospatial lead and manager of the project. As the geospatial lead, I developed the modeling framework to model the biophysical and socioeconomic impacts of sea level rise and enhanced storm surge caused by higher sea level and changes in land cover in both the natural and built environments. Specifically, I developed a subsidence rate grid that was used in sea level rise modeling, modeled socioeconomic impacts due to sea level rise, and designed the data-rich website with project results. There were two graduate students that I was coordinating and managing for the modeling work. As a project manager, I managed the project by coordinating and tracking a team of graduate students and research staffs along with two research groups in HRA.

9. **How was the project successful? What kind of impact has it had? Has this work been recognized for its significance, such as with awards or articles written about it? Has this work been used by other individuals or organizations, or changed others' practices in some way? Please provide specific examples where possible**.

It was a successful project as it was able to achieve all the objectives proposed to Grand Isle Endowment which funded this project, well-received in scientific community and by general public, and an additional component was also included that was not proposed in original proposal. So, in summary the objectives that were achieved are: mapping the current environment; modeling sea level rise and impacts on the natural and built environments; determining the economic impact due to sea level rise; analyzing the current laws and policies governing coastal zone management; developing a data rich website; and modeling the storm surge impact in current environments vs. future environments (an additional component).

I was covered by media regarding that project in May 2018. There are a couple of newspaper articles published in the front-page in Grand Isle Chronicle (May 4th, 2018) and Louisiana Express-News (May 7th, 2018) featuring me with my quotes. Both these newspapers have the highest circulation in the area.

The poster that I presented at 2016 Ocean Sciences Meeting is cited on Fourth National Climate Assessment (NCA4) which is a comprehensive and authoritative report on climate change and its impacts in the United States published every four years by U.S. Global Change Research Program.

The final deliverable of the project was a data and information-rich website that enables policy makers, managers and the general public to evaluate the impacts of future sea level rise on their communities with greater precision and accuracy.

Furthermore, this project led to additional funding from Louisiana Land Office for the Louisiana Coastal Resiliency Initiative study that I am going to talk about next.

1. **PROJECT 2 – Name**: Assessing Vulnerabilities of Sea Level Rise and Future Storm Surge Inundation for the Louisiana Coastal Resiliency Initiative
2. **Date project began**: September 2017
3. **Date project ended**: April 2019
4. **Please list any funding this project received (if applicable)**: $300,956 from Louisiana Land Office (LLO)
5. **Please list all resulting publications (if applicable)**:

GLO is going to incorporate my modeling work in the Louisiana Coastal Resiliency Initiative (LCRI) that is going to be made public either this week or next week. In addition to the plan itself, the 160 page technical report of all the modeling work that we did for the LCRI is also going to be available through GLO.

6. **Please provide any necessary background information for us to understand this project. What is the significance of this work? Why did you undertake this project? Was there a specific issue or problem you were trying to resolve? What were you hoping to accomplish**?

In March 2017, the Louisiana Land Office (LLO) completed phase one of the Louisiana Coastal Resiliency Initiative (LCRI) and our lab was also part of it. The 2017 plan describes the geological, ecological, and socio-economical condition of the coast and presents strategies and specific projects to improve resiliency, particularly in terms of improving the natural functioning of the coast. The 2019 plan is built from its predecessor but is also incorporating coastal modeling for future habitat change, flooding and storm impacts. The goal of these modeling work is to assist with identification of areas where resiliency or restoration project implementation would reduce future risk and costly damage. This is where my modeling work is used in the plan.

Wetlands on the Louisiana coast are particularly vulnerable to sea level rise due to low-lying topography, low tidal range relative to sea level rise, and in some regions, the presence of extensive coastal development acting as a physical barrier to the upland migration of wetlands as sea level rises. Furthermore, sea level rise has the potential to increase coastal flooding as storm surges of a given height are expected to occur more frequently. Therefore, the vulnerability of low-lying coastal areas on the Louisiana coast to flooding caused by storm surges will increase in the future. Therefore, it is important to identify the additional threat posed by storm surge and nearshore waves to people and the ecosystem. Thus, the modeling work is used in the plan to further comprehend and predict potential future scenarios for sea level rise, coastal storm and flood risk, and infrastructure impacts – all of which should be standard considerations for effective coastal planning.

The modeling work is used in the plan for a couple of reasons. First, the models are used to identify portions of the Louisiana coast most vulnerable to long-term processes, such as relative sea level rise and erosion. Coastwide assessments of these changes are conducted to identify future vulnerable areas. Second, the LLO is utilizing the models to determine if there is scientific evidence to defend the concept that large-scale coastal restoration projects are effective solutions for reducing the vulnerability of the Louisiana coast. To assess the effectiveness of these types of projects in Louisiana, a series of models are developed comparing the impacts from storm surge under two different scenarios: with modeled hypothetical large-scale restoration projects “with project” and if no action is taken “without project.” Model output from the two scenarios are compared to test whether the projects provided protection from surge both on the present landscape and the future landscape after decades of sea level rise.

7. **Please provide a plain language summary of what was done in this project. You should try to be detailed (2-4 paragraphs is appropriate), as more information will help us better understand the work**:

The study area of the “Living with Sea Level Rise on the Upper Louisiana Coast” project that I explained earlier was only the greater Grand Isle region. However, for this project, we modeled impact of sea level rise in 2100 of the whole 367 miles Louisiana’ Gulf Coast and more than 3,300 miles of bays and estuaries using the same modeling framework that I developed. Therefore, in this project, we used sea level rise, storm surge, and wave modeling to provide quantitative information regarding the impacts of relative sea level rise and enhanced storm surge caused by higher sea level and changes in land cover in Louisiana coast. For this project, the impact on natural and built environment due to one sea level rise scenario (1 m of global sea level rise by 2100) is modeled along the Louisiana coast, and six hypothetical Category 2 hurricanes that made landfall in different parts of the coast are modeled for both present and future landscape conditions to understand the different in storm surge impact in the present vs. future landscapes. Through modeling of these major hazards, this work shows the relative susceptibility to negative impacts on the natural and built environments along the coast. The work also simulates how the implementation of certain coastal resiliency projects and strategies may mitigate those impacts.

The LCRI recommends a list of high priority scientifically sound, feasible and cost-effective coastal resiliency projects based on the eight coastal issues of concern identified during the planning process in order to protect coastal infrastructure and natural resources. Therefore, the modeling work that we did to predict where future coastal hazards (sea level rise and storm surge) may impact Louisiana coast was used to better understand and defend the coast against those issues of concern. The modeling results help characterize how present-day built and natural environments are susceptible to climatic impacts, including relative sea level rise and coastal storm surge. These results not only support the need for present-day improvements, but also validate that the list of resiliency projects presented in the LCRI are viable solutions to the issues at hand.

8. **What was your specific role on this project? What were you individually responsible for? How was your work critical to the outcome of the project**?

I was directly involved on writing the proposal to Louisiana Land Office (LLO) to get funding of $300,956 for this project and I was listed as Lead Co-Investigator and a Primary Award Contact.

In regard to the modeling work, I am responsible for the storm surge modeling due to hypothetical Category 2 hurricanes making landfall in different regions of Louisiana coast in the present landscape and the predicted future landscape (2100 condition).

9. **How was the project successful? What kind of impact has it had? Has this work been recognized for its significance, such as with awards or articles written about it? Has this work been used by other individuals or organizations, or changed others' practices in some way? Please provide specific examples where possible**.

The modeling work for this project is completed and the results are used in the Louisiana Coastal Resiliency Initiative (LCRI). The modeling work is being used as a scientific evidence to defend and validate the list of resiliency projects presented in the LCRI as viable solutions to the issues at hand.

The LLO will be using the LCRI to emphasize the state's coastal needs and will submit it to the Louisiana State Legislature in the ongoing 2019 legislative session and request budgetary assistance to implement resiliency projects listed in it. Similarly, the LCRI can be used by federal, state and local decision makers to meet the restoration and planning needs of the Louisiana coast.

Furthermore, this project led to additional funding from Louisiana Land Office the Coastal Louisiana Study that I am going to talk about next, and just recently from the Port of Grand Isle.

1. **PROJECT 3 – Name**: Investigating sensitivity of storm surge elevation due to land cover changes caused by sea level rise
2. **Date project began**: January 2019
3. **Date project ended**: March 2019
4. **Please list any funding this project received (if applicable)**: $18,666 from Louisiana Land Office (LLO)
5. **Please list all resulting publications (if applicable)**: N/A

6. **Please provide any necessary background information for us to understand this project. What is the significance of this work? Why did you undertake this project? Was there a specific issue or problem you were trying to resolve? What were you hoping to accomplish**?

Along with the Louisiana Coastal Resiliency Initiative (LCRI), the Louisiana Land Office (GLO) in partnership with the U.S. Army Corps of Engineers (USACE) is also undertaking a planning study on Louisiana coast called the Coastal Louisiana Protection and Restoration Feasibility Study, also known as the Coastal Louisiana Study. The Coastal Louisiana Study will involve engineering, economic and environmental analyses on large-scale solutions to improve storm surge protection from hurricanes along the Louisiana coast, mainly the upper Louisiana coast in greater Grand Isle region. The final feasibility study and report will be completed in 2021.

In October 2018, USACE released the Draft Integrated Feasibility Report and Environmental Impact Statement (DIFR-EIS) that proposes a comprehensive plan to reduce damages from storm surge and restore the Louisiana coastal ecosystem through the combination of coastal storm risk management (CSRM) and ecosystem restoration (ER) alternatives. The CSRM alternative plans consist of structural features that include levees, floodwalls, surge barrier gates, and breakwaters to improve storm surge protection, especially in greater Grand Isle region. The ER alternative plans consist of nonstructural features that include habitat restoration and shoreline erosion control through marsh, beach/dune, and island restoration along Louisiana coast. All the CSRM and ER alternative plans were rigorously modeled based on economic, engineering, social, and environmental factors to identify the Tentatively Selected Plan (TSP). The TSP evaluation included a comparison of both the present and future conditions to improve storm surge protection for the coastal communities due to tropical storms and hurricanes, especially in greater Grand Isle region.

In order to model the future sea level rise conditions, the Coastal Louisiana Study just increased the sea level by certain amount in their storm surge models based on their sea level rise projection in the future but did not consider the land cover that is going to be changing due to sea level rise in the future. This is where our modeling results of TCRMP work (PROJECT 2 explained earlier) is applicable. For that work (PROJECT 2), we have modeled the projected future land cover changes due to 1 m global sea level rise in Louisiana coast using the currently available best input dataset. A handful of available studies that has looked into the effect of land cover (bottom friction) in storm surge modeling have shown that the bottom friction change (due to land cover changes) can play significant role on the hydrodynamics of storm surge modeling.

Therefore, this project provides quantitative information regarding the effect of land cover changes due to sea level rise on storm surge values. Basically, the purpose of this project is to understand the sensitivity of land cover changes (and thus bottom friction) on storm surge model outputs. I use the projected future land cover modeled for PROJECT 2 to calculate the bottom friction for the future condition storm surge modeling. This project incorporates storm surge modeling with three landscape scenarios: present landscape, future landscape with future sea level condition, and present landscape with future sea level condition. Three representative storms (Hurricane Ike and two synthetic storms) that pass through the greater Grand Isle region are modeled for all these three scenarios, therefore total nine storm surge modeling simulations are performed. The result of this project will be used by USACE for their future storm surge modeling work.

I undertook this project as sensitivity of land cover changes is an important consideration that need to be accounted for in any future condition storm surge modeling. Also, it was something that I was planning of doing in the near future for a manuscript that I am working on but I got funding for doing it. The result of this study will not only help refining the USACE modeling work but also my own modeling work in the future.

7. **Please provide a plain language summary of what was done in this project. You should try to be detailed (2-4 paragraphs is appropriate), as more information will help us better understand the work**:

The hydrodynamic storm surge modeling requires the physical system (real-world scenarios) to be accurately described and characterized in the given input files by providing the precise topographic and bathymetric elevations along with the land cover information, and also the meteorological inputs such as hurricane track, wind and pressure files. All storm surge models are strongly dependent on the accuracy of these input files, and the uncertainty or inaccuracy of these inputs will be reflected on the modeling results. As we say in Computer Science, garbage in, garbage out (GIGO) that means flawed input data produces flawed output.

Among many inputs to storm surge modeling, land cover is one of the important one. The hydrodynamic storm surge modeling requires an input of frictional roughness (bottom friction) that is characterized by the land cover over which wind blows, and wave and surge propagates. The speed at which the storm surge propagates is affected by land cover through bottom friction. Bottom friction is an important resistance mechanism that needs to be accurately quantified in the model. To analyze the effects of future landscapes and sea level rise on storm surge, we need to provide future land cover and topographic elevation to calculate the bottom friction in the future condition storm surge modeling.

The Coastal Louisiana Study, however, has used the future sea level rise condition but the present-day land cover and topographic elevation to calculate the bottom friction in their future condition storm surge modeling to identify the Tentatively Selected Plan (TSP) in the Draft Integrated Feasibility Report and Environmental Impact Statement (DIFR-EIS) that was released in October 2018. As a result, bottom friction change (due to land cover changes in future) does not play any role on the hydrodynamics that may contribute significant errors in the modeling results. Therefore, in this project, I use not only the future sea level rise condition but also the future land cover and topographic elevation outputs that I got from the PROJECT 2 (the LCRI project explained earlier) to understand the effect of land cover changes due to sea level rise on storm surge values. The results show that the future land cover change combined with sea level rise increases water levels and extent of inundation due to the same storm that makes landfall in the future in comparison to the present landscape. Therefore, this work will help U. S. Army Corps of Engineers (USACE) Galveston District, who is undertaking the Coastal Louisiana Study in collaboration with Louisiana Land Office, to understand how sensitive is the land cover changes on storm surge model outputs. This result will help them to evaluate their modeling work in the future.

8. **What was your specific role on this project? What were you individually responsible for? How was your work critical to the outcome of the project**?

I was directly involved on writing the proposal to Louisiana Land Office (LLO) to get funding of $18,666 for this project and I am listed as Lead Co-Investigator and a Primary Award Contact. I am the only person working on this project in close collaboration with USACE, from writing the proposal to the technical document that to be submitted to USACE after the completion of this project by the end of March 2019.

9. **How was the project successful? What kind of impact has it had? Has this work been recognized for its significance, such as with awards or articles written about it? Has this work been used by other individuals or organizations, or changed others' practices in some way? Please provide specific examples where possible**.

The main impact of this project is the USACE will be incorporating the result of this project in their future storm surge modeling work for the Coastal Louisiana Study. It has also opened up an opportunity of future collaboration with the USACE in the Coastal Louisiana Study that will be completed in 2021. I am told that more collaboration in modeling work will be forthcoming as this project shows that the land cover change results in additional inundation area and increase in water level during storm surge.

Section E: Professional Activities

This section is focused on activities outside of your educational and employment history. It can involve selective memberships or leadership positions with professional organizations, peer review and editorial board experience, work organizing professional conferences or seminars, etc. Try to think about any roles for which you were chosen based on the significance of your past work or your expertise in the field.

1. **Organization with which you were associated**: Journal of Coastal Research
2. **If known and notable, the reputation of the organization**: #4 ranked journal in Ocean and Marine Engineering (Google Scholar)
3. **What was your role, and why you were selected**? I am a reviewer for this journal and have reviewed one journal article so far. I was invited to review a manuscript in November 2018.
4. **Please provide a detailed description of your responsibilities and their importance to the organization**: Reviewing the manuscript submitted to Journal of Coastal Research

1. **Organization with which you were associated**: City of Grand Isle
2. **If known and notable, the reputation of the organization**:
3. **What was your role, and why you were selected**? I am a committee member of Watershore and Beach Advisory Committee appointed by the City of Grand Isle since 2015. I am appointed as an environmentalist in the committee based on my qualifications and experiences on coastal research by the City Council.
4. **Please provide a detailed description of your responsibilities and their importance to the organization**: I advise and make recommendation to the City Council on issues related to use or preservation of the waterfront, beaches and natural bodies of water within the city limits.

1. **Organization with which you were associated**: Central Coast GIS Users Group (CCGISUG)
2. **If known and notable, the reputation of the organization**:
3. **What was your role, and why you were selected**? I was a member of the organizing committee of 2012 Oregon Coastal Web Mapping and Geo-enabled Tools Meeting. The meeting was 1-day annual event with presentations related to mapping and geospatial technologies by researchers and practitioners in the Central Oregon Coast. I was selected due to my active participation in the Central Coast GIS Users Group (CCGISUG) while I was working for U. S. Geological Survey in Newport, OR.
4. **Please provide a detailed description of your responsibilities and their importance to the organization**: assisting the logistics of the meeting, communicating with researchers and practitioners for the talks in the meeting

1. **Organization with which you were associated**: 2009 Association of Computing Machinery (ACM) North Central Regional Programming Contest
2. **If known and notable, the reputation of the organization**:
3. **What was your role, and why you were selected**? I was a Judge in 2009 ACM North Central Regional Programming Contest which is the algorithmic programming contest for college students competing from universities in North Central North America (NCNA) region. The NCNA region includes Minnesota, Wisconsin, Iowa, North Dakota, South Dakota, Nebraska, Kansas, the Upper Peninsula of Michigan, and Western Ontario and Manitoba, Canada. I was selected as Judge because of my previous teaching in Nepal and at University of Nebraska-Lincoln, and computer programming experience.
4. **Please provide a detailed description of your responsibilities and their importance to the organization**: Judging the contest submissions by contestants and providing the final results to Regional Contest Director